Capstone 57 Research Paper: Modern TCP Congestion Control

Abstract

Transmission Control Protocol (TCP) governs reliable data delivery across the modern Internet. Traditional loss-based algorithms like TCP Reno interpret packet loss as the sole indicator of network congestion. However, in modern high-speed broadband networks with large buffers (bufferbloat), loss-based algorithms induce excessive queuing delays and throughput degradation.

Section 1: Comparative Analysis of Reno vs BBR

Google's BBR (Bottleneck Bandwidth and Round-trip propagation time) adopts a model-based paradigm. Rather than reacting blindly to packet loss, BBR dynamically probes and estimates the maximum bottleneck bandwidth and minimum round-trip time. In experimental 10 Gbps transatlantic testbeds, BBR achieves up to 4x higher throughput with 50% lower queue latencies compared to TCP Reno and CUBIC.

Section 2: Experimental Findings and Summary

In conclusion, model-based congestion control demonstrates superior fairness and efficiency across heterogeneous networks. For college computer networking students, understanding the mathematical trade-off between throughput and queuing delay remains a foundational concept.